جامعة عفت
EFFAT UNIVERSITY

Project Topic & Track Proposal

Real-Time Sign Language Recognition

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CS 4085: Deep Learning

Problem Statement & Alignment with Meta-Theme

Translating sign language in real-time traditionally requires heavy video processing, which is impractical for edge devices with constrained compute. This project strictly adheres to the "Tiny Deep Learning Models" theme by doing more with fewer resources. Instead of processing raw pixels, we propose using Google MediaPipe to extract 3D hand and body coordinate landmarks. We will train a lightweight model (such as a 1D-CNN or a small RNN) on these numerical arrays, heavily compressing the model size while maintaining fast on-device inference.

Project Objectives & Methodology

Core Objectives:

- **Data Processing:** Extract low-dimensional numerical features (hand landmarks) using MediaPipe to bypass heavy convolutional processing.
- **Architecture Design:** Train an efficient, lightweight neural network architecture suitable for constrained compute.
- **Optimization:** Apply model compression techniques, such as post-training INT8 quantization, to evaluate the accuracy/size trade-off.
- **Evaluation:** Report the exact parameter count, model size (in MB), and real-time inference latency alongside standard accuracy metrics.

Expected Outcome

A highly efficient, deployed model capable of classifying sign language in real-time via webcam. The final submission will demonstrate how coordinate-based processing and model quantization effectively minimize latency and memory footprint on standard hardware.